

Engineering Problem Solving Process

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Engineering Problem Solving Process

- Engineering problem solving provides a systematic way to generate and implement improvements
- It is especially useful when improving work methods, products, services, or systems
- Rather than jumping to one idea, engineers examine multiple alternatives and select the best one
- The process is iterative, evidence-based, and improvement-oriented

Why a Structured Process Is Needed

- Real-world engineering problems rarely have one obvious answer
- It is usually impossible to consider all conceivable alternatives at once
- A structured process helps narrow the search space and make rational choices
- The current method is also an alternative and should be evaluated against other possibilities
- Even an ideal future concept can serve as a useful reference point during solution search

Five-Step Engineering Problem Solving Process (DASSI)

- **Define** the problem
- **Analyze** the problem, including data collection
- **Search** for possible alternatives
- **Select** the best alternative through comparison
- **Implement** the solution and follow up

DASSI Engineering Problem Solving Cycle

Define the Problem

Clearly identify and understand the issue.



Step 1: Define the Problem

- A well-defined problem is already half solved
- Problem definition clarifies what is wrong, for whom, and under what conditions
- The key question is not merely “What can we improve?” but “What is the actual problem?”
- Poorly defined problems often lead to wasted effort and weak solutions
- A vague problem statement should be turned into a specific and actionable one

Why Problem Definition Matters So Much

- Good alternatives are often discovered already in the definition or analysis stage
- A problem must be real, meaningful, and worth solving
- There is no perfect work method; almost every system contains room for improvement
- Improvement opportunities may emerge because of changes in:
 - Quality requirements
 - Quantity or demand
 - Raw materials
 - Equipment
 - Time constraints
 - Organizational support

Step 2: Analyze the Problem

- Problem analysis requires careful observation and data collection
- The goal is to understand causes, constraints, and current performance
- Analysis should distinguish symptoms from root causes
- Data makes the improvement effort more objective and credible
- Without analysis, solution ideas are more likely to be based on assumptions

Step 3: Search for Alternatives

- Engineers should generate multiple alternatives before selecting one
- At least three alternatives should be explored whenever possible
- The existing method should be treated as one alternative
- A highly idealized or future-oriented concept may be another
- Additional alternatives can be generated through brainstorming and related ideation methods

Compare Alternatives and Select the Best One

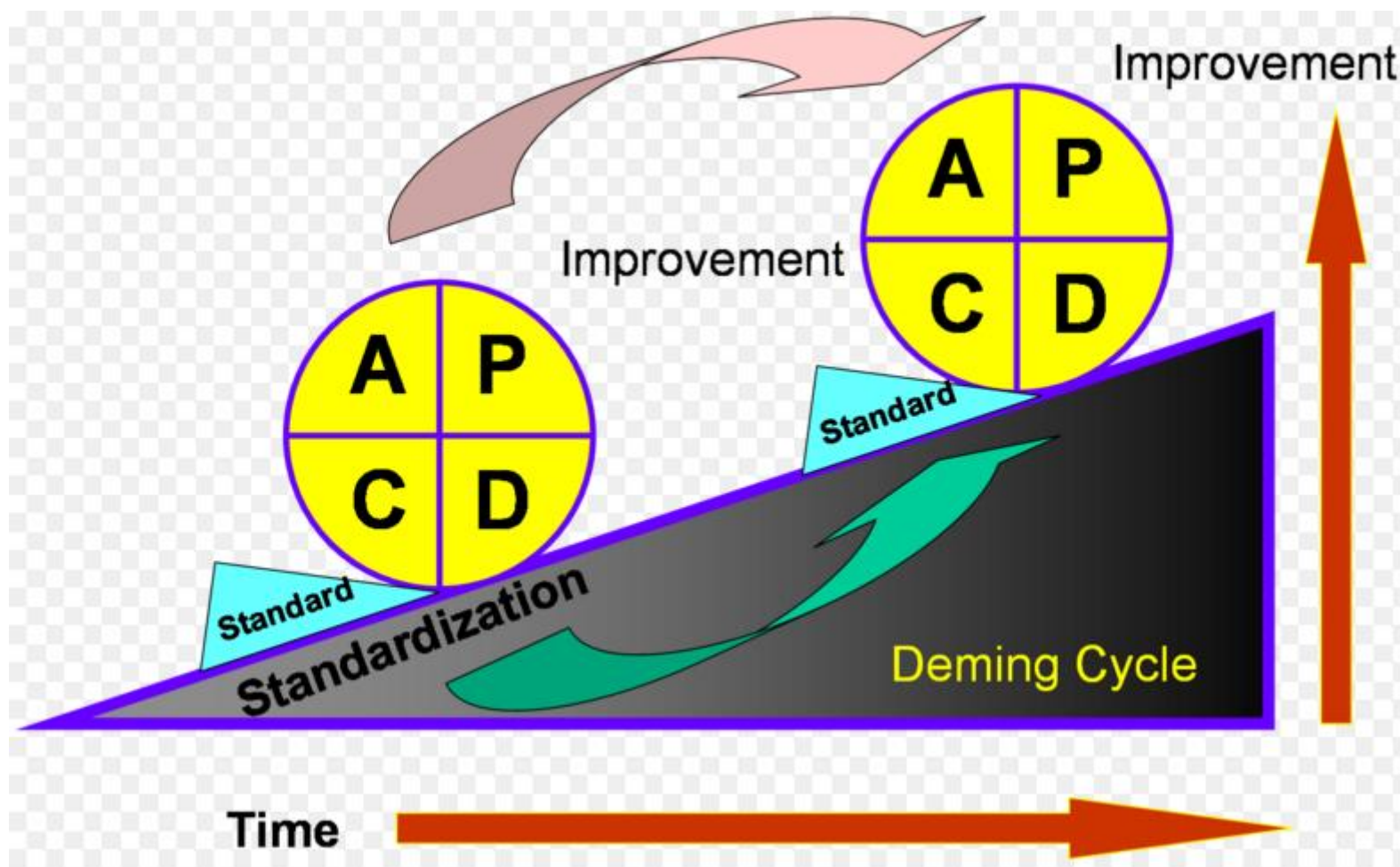
- Alternatives should be evaluated using explicit criteria
- Typical criteria include:
 - Effectiveness
 - Cost
 - Feasibility
 - Safety
 - Time
 - Compatibility with organizational conditions
- The goal is not to choose the most interesting idea, but the most appropriate one
- Selection should be evidence-based rather than intuition-driven

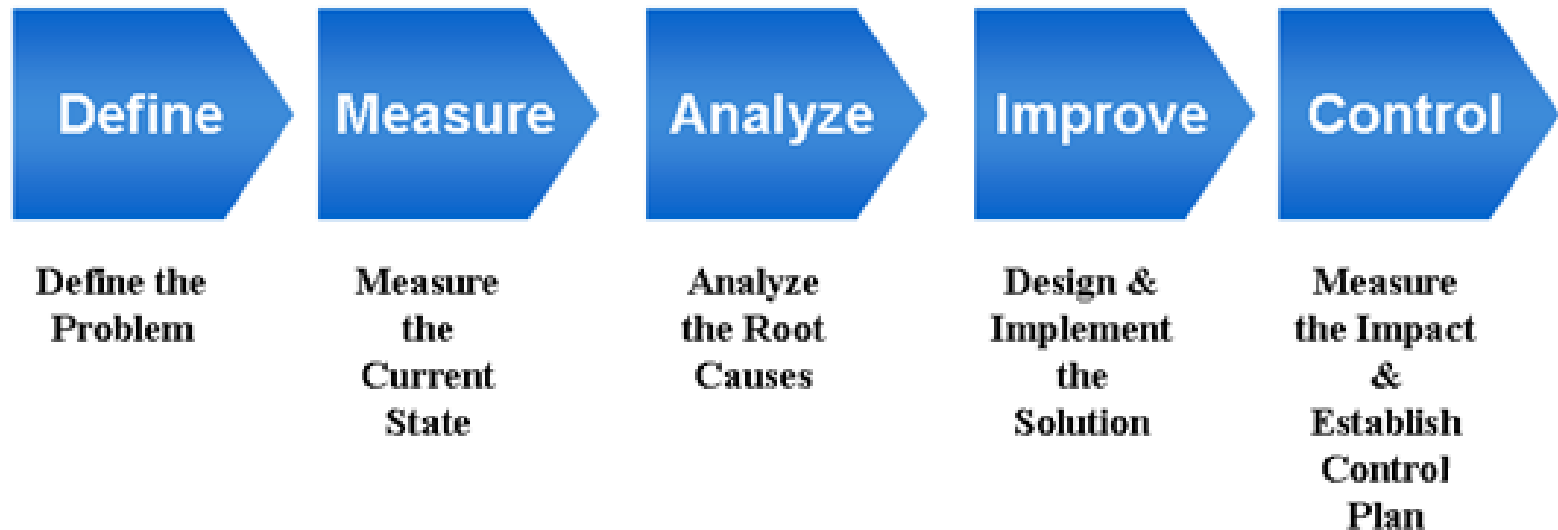
Step 5: Implement and Follow Up

- Improvement does not create value until it is implemented
- Follow-up is necessary to confirm whether the solution actually works in practice
- Implementation often reveals new issues not visible during analysis
- Continuous monitoring helps stabilize the improved system
- A solution is not complete until it is sustained

Related Improvement Cycles

- The engineering problem-solving process is conceptually similar to other improvement frameworks
- Examples include:
 - **PDSA**: Plan–Do–Study–Act
 - **PDCA**: Plan–Do–Check–Act
 - **DMAIC**: Define–Measure–Analyze–Improve–Control
- All of these frameworks emphasize structured improvement, evidence, and iteration
- The naming differs, but the core logic is highly similar





Key Principle: A Problem Well Defined Is Half Solved

- “A problem well defined is half solved already”
- “A problem identified is half solved already”
- These statements emphasize the value of disciplined thinking before action
- Many failed improvements originate from weak diagnosis rather than weak creativity
- Better understanding usually leads to better solutions

Why Good Ideas Often Fail

- Many good ideas are never adopted
- Technical merit alone does not guarantee implementation
- Resistance to change is one of the biggest barriers to improvement
- In practice, an organization may reject useful ideas for social or emotional reasons
- Engineering improvement therefore involves both technical and human challenges

Resistance to Change: Why It Happens

- People tend to prefer familiar methods and routines
- A new idea may be interpreted as criticism of the current method
- People may resist ideas proposed by others, especially outsiders
- Fear is often involved:
 - Fear of failure
 - Fear of loss of status
 - Fear of disruption
 - Fear of embarrassment
- Past frustration may also matter if earlier good suggestions were ignored

How to Reduce Resistance to Change

- Reduce fear by communicating clearly and respectfully
- Build teamwork around the improvement effort
- Involve the people affected by the change
- Create trust before demanding adoption
- People are more likely to accept change when they feel respected and included
- Implementation is easier when improvement is seen as shared progress rather than imposed criticism

Other Barriers to Adoption

- Even beyond psychological resistance, practical barriers can block implementation
- Important considerations include:
 - Risk factors
 - Training needs
 - Initial trial and startup issues
 - Maintenance and repair issues
 - Timing
 - Safety
 - Company policy
- A technically good solution may still fail if these realities are ignored

Final Takeaways

- Engineering problem solving is systematic rather than accidental
- Good improvement begins with clear problem definition and careful analysis
- Multiple alternatives should be generated and compared
- Implementation and follow-up are essential parts of the process
- Human factors, especially resistance to change, must be managed along with technical factors
- Effective improvement requires both sound logic and organizational sensitivity